

AAI201 MACHINE LEARNING CAPSTONE

Predicting Match Outcomes in Age of Empires II

Problem Statement & Goals

The Challenge

Competitive AoE2 players often rely on anecdotal experience. Identifying which civilizations or behaviors truly influence outcomes requires a rigorous quantitative approach.

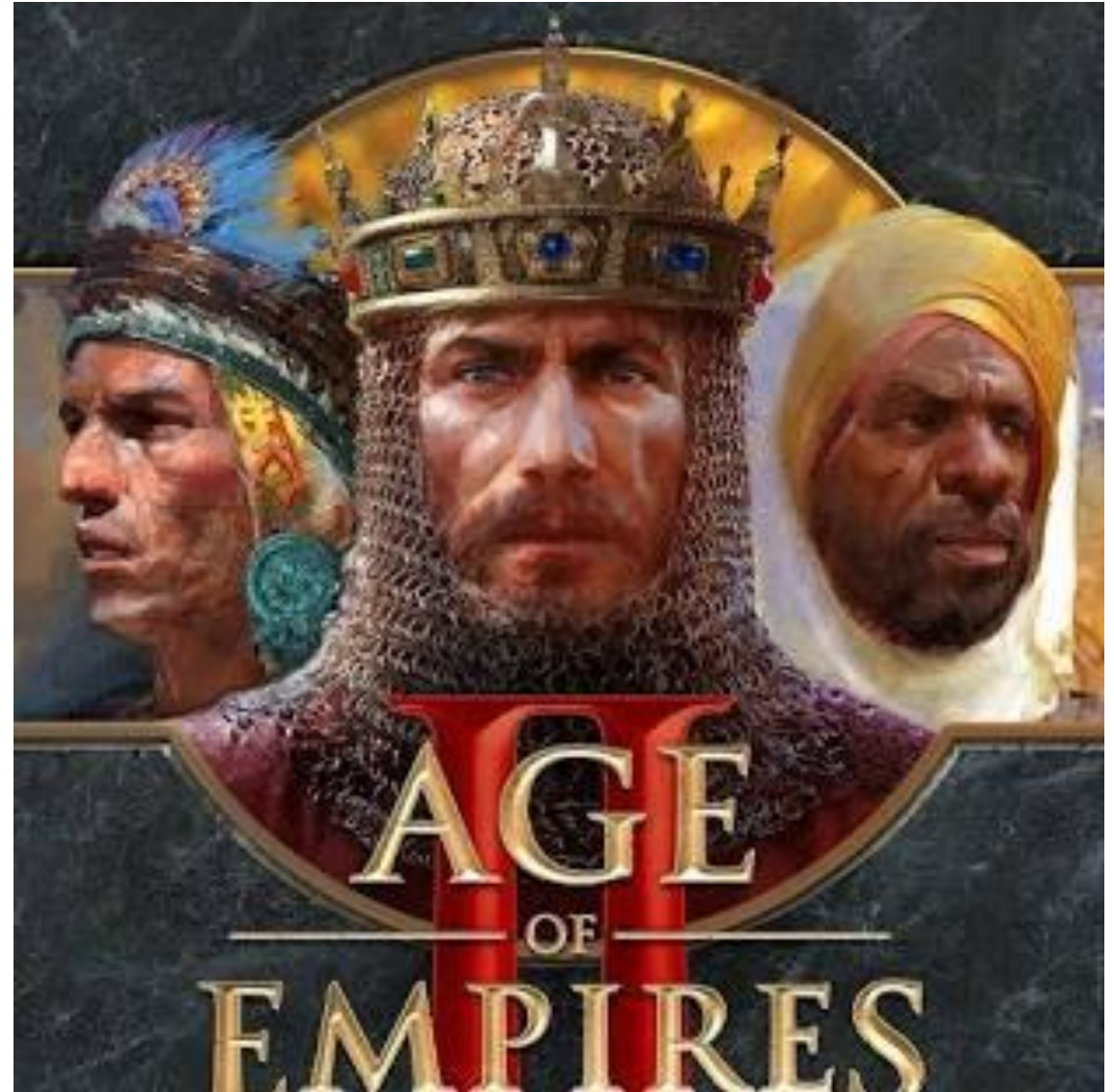
Primary Objectives

Build a predictive model to estimate win probability and quantify the "Civilization Advantage" across different map types and player skill brackets.

Dataset Analysis

Kaggle AoE2 Match Data

The study utilized a comprehensive dataset of over 100,000 ranked 1v1 matches. Key features include Player Elo, Civilization choice, Map Type, and Match Duration. Preprocessing involved handling categorical variables and filtering for balanced competitive play to reduce noise.



Tools and Algorithms



Core Stack

Python with Pandas for data manipulation and Scikit-Learn for modeling.



Random Forest

Chosen for its ability to handle categorical data and non-linear relationships.



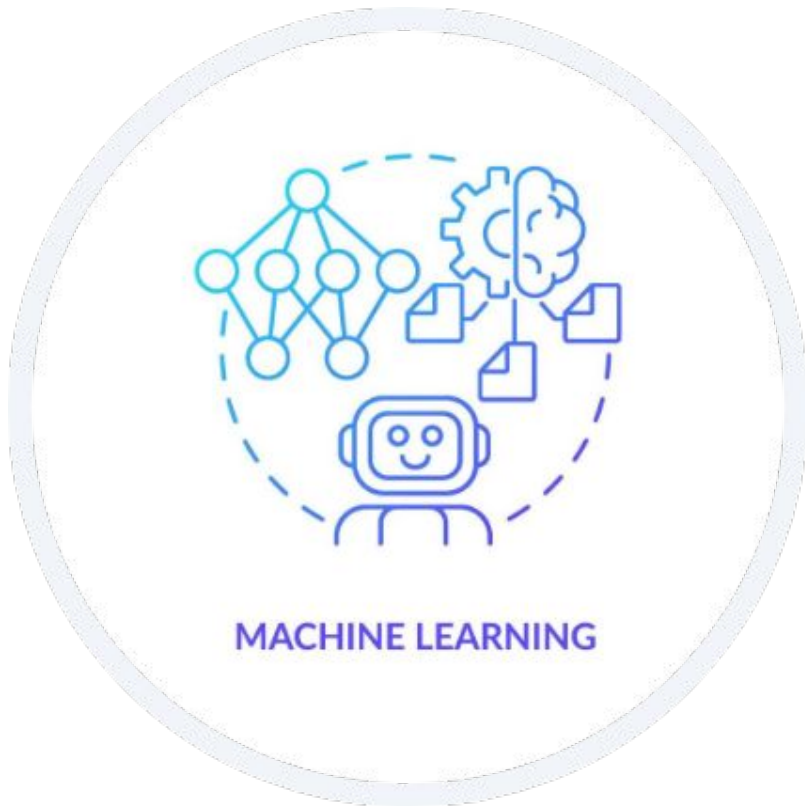
Encoding

One-Hot Encoding was applied to 40+ civilizations to maintain mathematical integrity.

Overfitting Prevention

- ✂ **Model Pruning:** Implementation of a strict `max_depth=5` to prevent trees from memorizing noise.
- ✎ **Validation Strategy:** A 80/20 train-test split ensured the model was evaluated on previously unseen match data.
- 📄 **Complexity Control:** Utilized `min_samples_split` to focus on broad strategic patterns rather than niche cases.

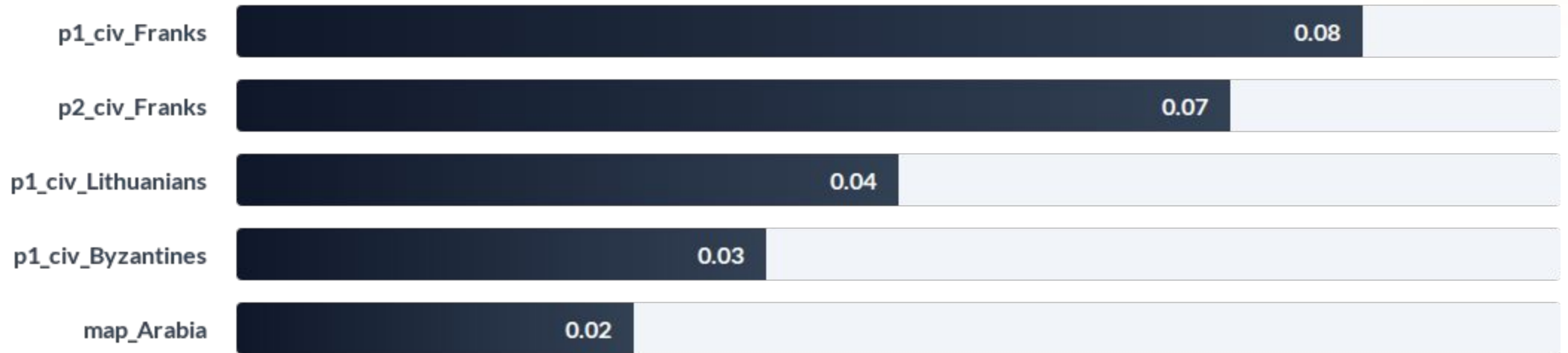
Bias & Fairness



Addressing Data Leakage

Initial results showed 100% accuracy due to post-match "Winner" metadata. I implemented a random shuffling algorithm to balance player positions and removed post-game features, ensuring a fair strategic analysis.

Top Factors Influencing a Win



Relative Importance of Strategic Features

Model Performance

52.15%

Realistic Accuracy

The Strategic Edge

In a highly balanced game, a 52% predictive accuracy based purely on pre-game choices (Civilization and Map) proves that strategy provides a quantifiable advantage even before the first unit is produced.